

Fifty-State Hosting Plan

Where to host a platform of fifteen FastAPI services, PostgreSQL + PostGIS, and S3-stored raw lab payloads — starting at 1–2 states, built to reach all 50 with redundancy, automated backups, mirroring, and load balancing.

Recommendation: AWS

Run **RDS for PostgreSQL (Multi-AZ)** with PostGIS, **native S3** for lab payloads, and the fifteen services on **EC2 under systemd behind an Application Load Balancer**. Every requirement on the list — point-in-time recovery, cross-region mirroring, load balancing, HIPAA-grade compliance, and a 50-state growth path — is a configuration step on AWS, not a migration.

Why AWS specifically

- **The code already speaks S3 — AWS is where that dialect is native.** No compatibility layer, no signature quirks. Versioning, lifecycle tiering, and Cross-Region Replication (your "mirroring") are bucket settings, not projects.
- **Managed Postgres with PostGIS, exactly as specified.** RDS for PostgreSQL ships PostGIS as a supported extension, with automated daily backups plus point-in-time recovery (up to 35 days) out of the box. Multi-AZ gives synchronous standby failover; read replicas and Aurora Global Database are the same engine when scale demands them.
- **Day one looks exactly like the plan you already wrote.** An EC2 instance running fifteen systemd units is your current design, verbatim — so the hosted flip changes one variable (where it runs), not two (where it runs *and* how it's orchestrated). ECS Fargate is available later without changing providers.
- **Compliance that survives 50 state contracts.** Raw lab payloads are very likely PHI: AWS signs a HIPAA BAA covering RDS, S3, EC2, and ALB, and holds

FedRAMP authorizations. GovCloud (US) exists if a state contract ever demands it. State procurement offices already know AWS paperwork.

- **The most US regions of any provider.** us-east-1, us-east-2, us-west-1, us-west-2, plus two GovCloud regions — enough for genuine geographic redundancy and any residency clause a state slips into a contract.

How the field compares

PROVIDER	MANAGED POSTGIS	S3 API	HIPAA BAA	US REGIONS	VERDICT
AWS	Yes — RDS / Aurora	Native	Yes	4 + 2 GovCloud	Recommended
Google Cloud	Yes — Cloud SQL	Interop mode, imperfect	Yes	~12 (smaller footprint each)	Solid runner-up
Azure	Yes — Flexible Server	No — Blob API differs	Yes	Many	Breaks the S3 design; only if a state mandates Microsoft
DigitalOcean	Yes — Managed PG	Spaces (S3-compat)	No	3	Cheapest start, but no BAA — disqualifying if payloads are PHI
Hetzner / OVH	Partial	S3-compat storage	No	1-2	Cost play only; too thin for this

"Interop mode" on GCP: Cloud Storage accepts S3-style requests but with enough auth and semantics differences that "the code already speaks S3" stops being free.

BEFORE THE FIRST REAL PAYLOAD

If lab payloads contain PHI, sign the AWS Business Associate Addendum (a self-service click-through in AWS Artifact) **before** production data lands, and turn on encryption at rest (default on RDS and S3) plus CloudTrail audit logging. Retrofitting compliance is far more expensive than starting with it.

Rollout in three phases

States are *tenants*, not regions — adding a state is a data and onboarding event, not an infrastructure build. The infrastructure only steps up three times.

PHASE 1

1-2 states · one region

≈ \$400–700 / mo

- **RDS PostgreSQL Multi-AZ** with PostGIS — automated backups, 35-day point-in-time recovery, synchronous standby in a second availability zone.
- **S3 bucket with versioning** for raw lab payloads; lifecycle rules tier old payloads to Glacier.
- **Two EC2 instances behind an ALB**, each running the fifteen FastAPI services as systemd units (provisioned by cloud-init or Ansible). Two boxes instead of one is the entire redundancy story at this scale.
- **Route 53** for DNS with health checks; ACM for free TLS on the load balancer.

PHASE 2

3-15 states · hardened + mirrored

≈ \$1.5k–4k / mo

- **Terraform everything** — from here on, infrastructure changes are code review, not console clicks.
- **S3 Cross-Region Replication** to a second region: the mirroring requirement, live.
- **RDS read replica** (in-region for load, cross-region for disaster recovery) plus AWS Backup vault copies to the second region.
- **Auto Scaling group** for the service tier, or migrate the fifteen units to ECS Fargate if per-service scaling starts to matter.
- **CloudWatch alarms, Secrets Manager, WAF** on the ALB — the operational floor for handling many states' data.

PHASE 3

Toward 50 states · multi-region

≈ \$8k–20k+ / mo

- **Aurora PostgreSQL** (drop-in from RDS) with **Global Database** — storage-level replication to a second region, sub-minute failover.

- **Active/passive multi-region:** full stack standing in region two, Route 53 failover routing in front. Go active/active only if latency data ever demands it.
- **Partitioning decision:** one Aurora cluster with state-partitioned tables usually beats 50 small databases — revisit with real per-state volumes.
- **CloudFront** in front of the ALB for TLS termination at the edge and DDoS absorption.
- **GovCloud enclave** only if and when a specific state contract requires it.

THE ONE-SENTENCE VERSION

Start with the smallest honest AWS footprint — Multi-AZ RDS, versioned S3, two EC2 boxes behind an ALB — and let backups, mirroring, and multi-region redundancy arrive as configuration on the same platform, so scaling to 50 states never requires changing providers or rewriting the S3-speaking code.

Costs and cautions

Figures above are planning bands, not quotes — RDS instance class and payload volume dominate. Two habits keep AWS bills sane: tag every resource from day one, and set an AWS Budgets alarm before the first deploy. The main lock-in risk is Aurora-specific features; staying on vanilla PostgreSQL semantics and the plain S3 API (both true of this design) keeps the exit door open.

If the payloads turn out to carry no PHI and budgets are tight, DigitalOcean is a legitimate Phase-1 shortcut (managed Postgres with PostGIS, S3-compatible Spaces, load balancers) — but it has no BAA and three US datacenters, so it is a place to start, not a place to reach 50 states.